Dr. Parth **Nayak**

Data Scientist | AI Researcher | Astrophysicist

- Munich, Germany
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- par-nay

Education

- **Ph.D. in Astrophysics** (Dr. rer. nat.)
  LMU Munich | Dec 2025
- **B.Sc.+M.Sc. in Physics** (dual degree)
  IISER Bhopal | Jul 2020

Skills

AI / Data Science

Deep machine learning (CNNs, MLPs, Transformers), simulation-based inference (SBI), evolutionary algorithms, Gaussian processes, Bayesian & Monte Carlo methods, density estimation

Programming

Python, LaTeX (proficient); HTML, CSS, Javascript (intermediate); C++, Java (basic)

Python packages

NumPy, SciPy, Pandas, Matplotlib, TensorFlow, Keras, Optuna, H5Py, AstroPy, HealPy, *among others*

OS / Shell

bash, Z shell (proficient), PowerShell (basic)

Other IT

High-performance computing (HPC), Git version control, LLM prompt engineering, Markdown, Mac/MS Office apps

Languages

English (full working proficiency); Gujarati, Hindi (native/bilingual); German (B1/limited working proficiency); French (beginner)

Highlights

- **5+ years** of **research experience** in **Data Science** and AI algorithm development
- **Led three and coauthored one** research papers on novel AI applications in astrophysics
- Delivered **10+ technical talks** at **international conferences** and **research institutes**
- Received **5+ academic awards** and **fellowships**

Relevant experience

Doctoral Candidate: Astrophysics & Data Science *LMU Munich | Oct 2021 - Dec 2025*

- **Developed a SBI framework "*LyaNNA*"** [par-nay/lyanna] based on a novel 1D ResNet with Python/Keras from scratch that **outperforms** conventional methods by **>100%** in inference precision for complex, realistic, noisy spectroscopic data
- **Developed "*SynTH*"** [par-nay/synth] to model intergalactic absorption from massive cosmological **hydrodynamic simulations** and synthesize training data for *LyaNNA*
- **Led two and coauthored one** research papers on AI for inference in astrophysics [Nayak+ 2024, 2025, Chang+ 2025]
- Conducted research at the **Hightech Agenda Bayern Chair of Astrophysics, Cosmology & AI** (ACAI) on AI applications in cosmology

Master's Thesis: Astrophysics & Data Science *IISER Bhopal | Aug 2019 - Jul 2020*

- **Developed a GA framework "*GA-ILC*"** to optimize the reconstruction of cosmic microwave background (CMB) from contaminated observations
  - A novel evolutionary algorithm [par-nay/genepy] to extract faint cosmological signals from high-noise environments (foreground removal)
- **Published** one research paper on evolutionary algorithms in astrophysics [Nayak & Saha 2022]

Leadership & supervision

- **Co-supervisor of LMU Master's theses and labs** *Oct 2021 - Dec 2025*
  Formulated and supervised two LMU Master's theses, and co-supervised 5+ computational and instrumental lab courses
- **Co-organizer of "Astronomy on Tap" (AoT) Munich** *since Oct 2025*
  Coordinating the monthly outreach event to engage the general public with accessible scientific talks and pub-quizzes
- **Co-organizer of "Meet the Speaker" @LMU Observatory** *Oct 2023 - May 2025*
  Revitalized the weekly series by relocating it to the historic 200-year old telescope dome, a unique atmosphere for students for informal networking with international experts
- **Coordinator & Senior Advisor of IISER Bhopal Astronomy Club (IBAC)** *2016 - 2020*
  Played a foundational role as the club's second coordinator, establishing enduring traditions and growing the community into one of India's most popular student astronomy clubs
- **Physics Department Representative** *Aug 2017 - Jul 2018*
  Unanimously elected to serve as the primary liaison between faculty and a class of 60+ physics major students

Selected awards & fellowships

- **DAAD Research Grant — Ph.D. scholarship** *Germany | Oct 2021 - Dec 2025*
- **Proficiency Medal in Physics — IISER Bhopal** *India | Jul 2020*
- **DAAD-WISE Scholarship** *Germany | May - Jul 2019*
- **INSPIRE Scholarship for Higher Education** *India | Aug 2015 - Jul 2020*